Quechua-Spanish Translation & Optimization of Tokenization for Agglutinative Languages

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Abstract

Machine translation for low-resource agglutinative languages remains challenging because standard tokenization methods often fail to capture meaningful morphological structure. We investigate whether linguistically informed tokenization can improve Spanish-Quechua translation by replacing byte-pair encoding with morpheme-based segmentation from a finite-state transducer (FST). Starting from Meta's NLLB-200 distilled 600M parameter model, we extend the tokenizer's vocabulary with morpheme tokens and post-train the model on the Somosnlp Hackathon 2022's Spanish to Quechua dataset. We evaluate two tokenization variants using BLEU, chrF, and chrF++ metrics. Our best performing model outperforms the baseline achieving a BLEU score of 10.7 (vs. 3.0), a chrF score of 43.3 (vs. 30.1), and a chrF++ score of 38.3 (vs 25.6), demonstrating that morphologically aware tokenization substantially improves translation quality for Quechua. These results suggest that incorporating linguistic structure into tokenization is a promising strategy for improving neural machine translation in low-resource agglutinative languages.

1 Introduction

Low-resource languages have historically been more difficult to translate accurately due to the scarcity of available data. In this project, we aimed to improve translation despite the lower amount of resources by proposing an alternative method of tokenization for Quechua, the most spoken indigenous language in South America. Tokenization is a fundamental process in natural language processing that breaks words into smaller units to better capture their distribution. Instead of standard byte-pair encoding, we segmented tokens by linguistically determined morphemes (segments of a word with independent meaning). As an agglutinative language, Quechua has many more morphemes per word than is typical for English, Mandarin, or other high-resource languages, which were used to train the most developed NLP models. We hypothesized that improving the morphological alignment of tokens will improve translation.

2 Literature Review

Jinfan Frank Hu (2025) previously explored how different tokenization strategies performed for low-resource agglutinative languages in the context of Turkish and Finnish, testing byte-pair encoding (BPE), word-level, character-level, and n-gram tokenization, and determined that word-level tokenization performed better than other tokenization strategies, suggesting that preserving linguistic structure was more effective than statistical frequency-based segmentation.

Since much more information can be contained in a single word in agglutinative languages than in fusional languages like English, many more unique words are used/created, so full word-level tokenization requires much more data to have the same quantity of observations per word.

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"Analyzing BPE output across morphologically rich languages, we observe that its segmentation often disregards meaningful morpheme boundaries, introducing ambiguity and disrupting semantic coherence" (Asgari et. al., 2025)

MorphBPE attempted to integrate linguistic information into the byte-pair encoding process and succeeded in achieving better performance when training LLMs. It was evaluated using morphological edit distance and training cross-entropy loss compared to standard BPE, but required further extrinsic evaluation of LLM performance. We used BLEU (Papineni et al., 2002), chrF (Popović, 2015), and chrF++ (Popović, 2017) metrics to evaluate the performance of translation using linguistically informed tokenization.

3 Methods

The model that we used for our baseline is Meta's NLLB-200 distilled 600M parameter model (Facebook/Nllb-200-Distilled-600M · Hugging Face, 2022) originally trained on over 200 languages, including our target language, Quechua. In addition, we are using Meta's BPE tokenizer released alongside NLLB-200 as the tokenization strategy for our baseline. We used a pretrained model rather than building our own because we lack sufficient computing power and data to confidently build an effective baseline that would allow us to focus on tokenization metrics rather than just translation accuracy (which has been explored repeatedly).

To evaluate our base model, we compared the model's machine translation of Spanish to Quechua against reference translations. Specifically, we will use BLEU, chrF, and chrF++ as our evaluation metrics. BLEU is a machine translation evaluation metric that compares the n-gram distribution of words in generated text to the n-gram distribution of words in the reference text, and chrF is a machine translation evaluation metric that compares the n-gram distribution of characters in generated text to the n-gram distribution of characters in reference text. chrF++ is another machine translation evaluation metric that combines BLEU and chrF to compare the distributions of the n-grams of words and characters in generated text to reference text. To compute these scores, we used the SacreBLEU package (Post, 2018).

Our tokenization was performed using a modified version of a Quechua Finite-State Transducer (FST) developed by Annette Rios. FST's are a linguistic tool that converts from surface forms of a word (words as written) to the underlying morphemes, such as grammatical case, plurality, or tense. It was modified to maintain the splitting of the word into its morphemes, but maintain the original suffix i.e. instead of converting paykuna→3rdPersonSingular+Plural, it converts paykuna→pay+kuna.

We extended the vocabulary of the base model with these new tokens, initialized them with average embeddings that would've been produced by the base NLLB, and then post-trained the model using Somosnlp Hackathon 2022's Spanish to Quechua dataset (Spanish-To-Quechua, 2022) which contains just over 100k samples for training.

We trained one model with word initial morphemes tokens (WIM) where each quechua morpheme produced from the FSTs decoding that could be word initial, roots and parts of speech, had duplicate variants of those morphemes representing the start of the word. This aligns with our baseline model since the baseline model implementation defaults to representing spacing between words with morphemes that have '▁' (SU) as its prefix. Note that this character is different from the standard underscore keyboard '_'. We also trained another model without word initial morphemes (ND - No Duplicates) where spacing between words was encoded by adding lone SU tokens at word boundaries.

To encode our reference texts using our FST we first split by whitespace getting text chunks and then split those text chunks via regex into chunks of two kinds: word candidate chunks (containing alphabetic characters and apostrophes with at least one alphabetic character) and non-word chunks (all other characters including numbers and punctuation). Only word candidate chunks are processed by our FST. Across our train, validation, and test sets, there are 1.40 million text chunks of which 1.07 million are word candidate chunks. Out of these word candidate chunks 566 thousand are successfully encoded with our FST (40.4% of all chunks).

4 Results

On Somosnlp Hackathon 2022's Spanish to Quechua test dataset (Spanish-To-Quechua, 2022) which contains almost 13k samples, the baseline NLLB-200 distilled 600M parameter model achieved a BLEU score of 3.0, a chrF score of 30.1, and a chrF++ score of 25.6.

After post-training with our FST morpheme tokenization, the ND model significantly outperformed this baseline. Against the original reference translations, its decoded output achieved BLEU, chrF, and chrF++ scores of 10.7, 43.3, and 38.3, respectively. Even when evaluating with the raw model outputs against the reference text, our ND model still outperformed the baseline on chrF and chrF++ although it underperformed on BLEU. In this context, we expected our models to do worse on BLEU since the raw model outputs contain FST tags which artificially causes word level mismatches and drives the BLEU score down; the chrF and chrF++ scores remain high because chrF rewards the model at the character level rather than the word level. We view the decoded vs original reference as a more meaningful comparison since it removes the FST prefix tags which are an artifact of our tokenization method rather than a property of the underlying translation.

While the WIM model performed worse than the ND model across all three metrics, it still outperformed the baseline on every metric when comparing its decoded output to the reference text. Given that the Somosnlp Hackathon 2022's Spanish to Quechua training set is relatively small for machine translation, we suspect that this gap stems from the duplicate root and part of speech morphemes splitting the training signal across two embeddings without enough data to converge to comparable performance.

Table 1: Sample Spanish → Quechua translations generated by baseline, ND, and WIM models with greedy decoding.

Source (Spanish)	Reference (Quechua)	Baseline	ND	WIM
Como ser humano perfecto, Jesús podía ofrecer un rescate.	Jesusqa mana pantaq kaspanmi "llapallanchikta salvawananchikpaq" wañunan karqa.	Payqa, Diospaq allin kajtinqa, qespichisqa karqa.	Part_Neg pantaq runa kasqanraykum Jesusqa wañuyta atirqa.	Jesusmi Part_Neg pantaq runa kaspanqa, wañuyta atirqa.
Somos criaturas físicas hechas del polvo.	Allpamanta rurasqa runam kanchik (Genesis 2: 7; 3: 19).	Chayraykun, laq'amanta ruwasqa kayniyoq kayniyoq kayniyoq kayniyoq*	allpamanta rurasqa kasqanchikraykum.	allpamantaqa allpamanmi allpamanta ruwaspami.

*The baseline continues repeatedly generating ' kayniyoq' until the max length is reached.

Table 2: Evaluation metrics (BLEU, chrF, chrF++) for the ND and WIM models comparing model output (raw) and FST-prefixed-removed output (decoded) against the original reference text.

Model	Output Type	BLEU	chrF	chrF++
ND	raw	1.867	31.810	27.809
ND	decoded	10.717	43.348	38.263
WIM	raw	1.282	23.241	20.199
WIM	decoded	6.843	33.081	28.904

* The decoded output removes the "+" and "=" characters that represent the inter-morpheme and start of word tokens from the FST, respectively. It makes sense that the decoded outputs are lower-scoring in the second table, as passing the translations through the FST effectively "encodes" them, but a bit more faithfully as the decoding is not a 1 to 1 process.

Table 3: Evaluation metrics (BLEU, chrF, chrF++) for the ND and WIM models comparing model output (raw) and FST-prefixed-removed output (decoded) against reference text passed through the FST.

Model	Output Type	BLEU	chrF	chrF++
ND	raw	33.985	46.997	43.761
ND	decoded	15.920	46.110	41.353
WIM	raw	27.958	37.526	33.239
WIM	decoded	10.613	35.909	31.886

5 Discussion and Analysis

Our model scored significantly better with chrF compared to BLEU. This is expected since Quechua is an agglutinative language and BLEU is a word level metric while chrF is a character level metric. Quechua being an agglutinative language means phrases that would be multiple words in many non-agglutinative languages are often instead one long word of concatenated morphemes. Thus, we would expect that our model's performance on BLEU to be relatively worse than its performance on chrF since a character level evaluation gives partial credit for generated words that have a mix of correct and incorrect morphemes while a word level evaluation does not. This can be seen in Table 1, example 1, where we observe some similarity from outputs despite only sharing the word "pantaq" and the "Part_Neg" part of speech tag across translations. BLEU would give zero credit for generating "wañuyta" (ND) and "kaspanqa" (WIM) (instead of "wañunan" and "kaspanmi") whereas chrF would assign some value to correctly generating the character sequences "wañu" and "kaspan". This is preferable, as the only difference between kaspanqa and kaspanmi is "-qa", a suffix that acts as a topic marker, and "-mi" a suffix denoting evidentiality. It does not change the entire meaning of the word, like the words "sever" vs "seven" in English.

The baseline model had a chrF++ score of 25.6 on the test dataset, and by post-training using our custom tokenization and decoding the output, our ND model achieved a chrF++ score of 38.263 when comparing to reference translations that had not been passed through the FST. The score is even higher when comparing the raw output from the model to the reference translations that have been passed through the FST to decompose them into their morphemes, likely due to the FST being unable to keep certain parts of speech as their surface forms. For example, see the translation below, with [ES] as the original Spanish (model input), [BASE] as the original baseline model without modification to vocabulary or post-training, [TRND] as the decoded output from the trained version of the model (WIM), and [REF] as the reference translation.

[ES] SIN duda alguna, la Biblia no tiene rival en la historia.
[BASE] Bibliaqa mana iskayrayasqachu kay tiempopi.
[TRND] Part_Affir, Bibliapiqa manam PrnInterrpas tupanchu.
[REF] ARÍ, enteron pachapiqa manam ima libropas Biblia hinaqa kanchu.

In the trained FST output, PrnInterr is an interrogative pronoun and Part_Aff is an affective particle. Certain qualities were harder to modify in the FST for a few of these underlying forms to have them maintain their surface form, likely due to the lack of hard-coded one-to-one correspondence. This will reduce the scoring for the decoded evaluation metrics as compared to the encoded evaluation metrics for the reference outputs that were run through the FST as well.

Using the ND model that was more faithful to original model encoding (including start-of-word tokens), we achieved a final training loss of 1.12, and a final test loss of 1.48. The WIM model that dropped start-of-word tokens achieved a final training loss of 2.46 and a final test loss of 2.60, even after more epochs.

Across every evaluation metric, both at word-level using BLEU and character-level using chrF, the ND model performed the best when comparing to reference translations that had been run through the FST. As seen in Table 2 in the results section, this also holds true when using the decoded ND output vs the raw reference translations, which is likely the best metric to use when extrapolating results from this training, as it doesn't depend on the FST for both the inputs and the output. We find that using linguistically informed tokenization does significantly improve the accuracy of translation for Quechua.

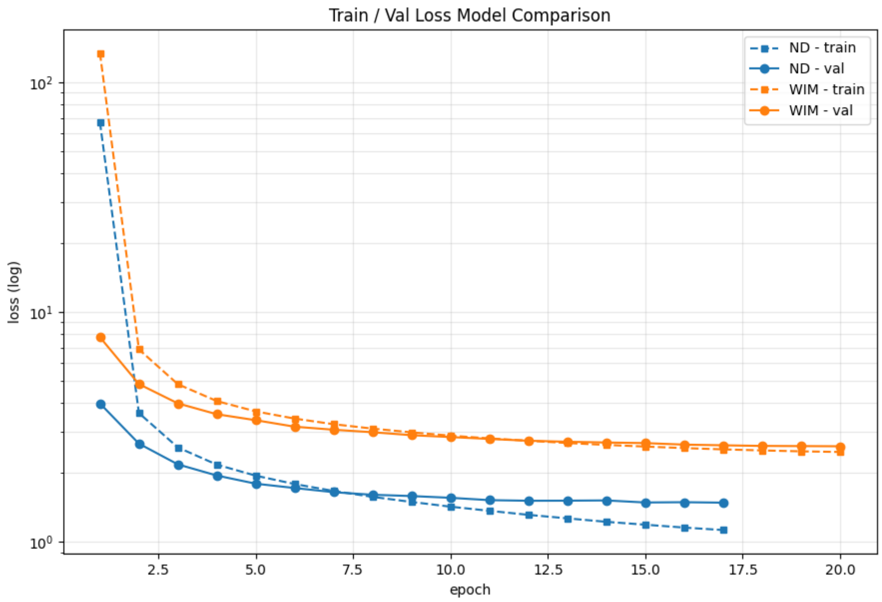

Figure 1: Train vs Test Loss Across Models Over Epochs

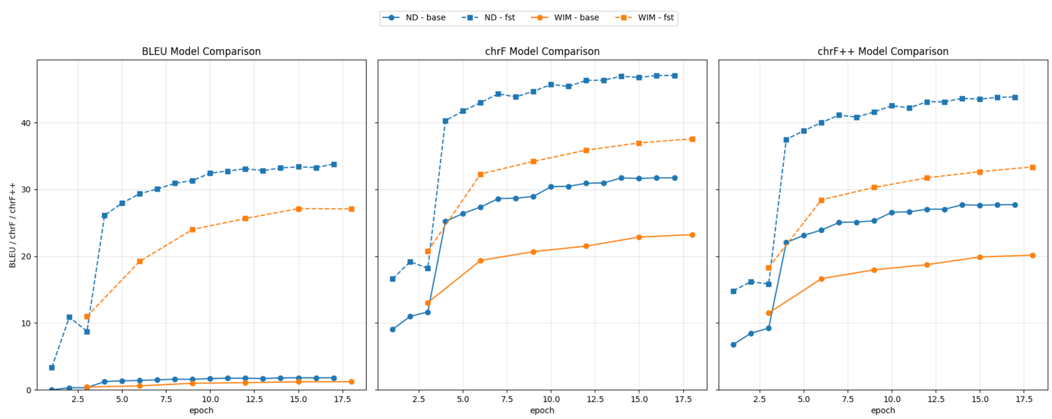

Figure 2: Evaluation Metrics Across Models vs FST-Processed Reference Output

References

N. Ahmed, N. Flechas Manrique, and A. Petrović. Enhancing Spanish-Quechua machine translation with pre-trained models and diverse data sources: LCT-EHU at AmericasNLP shared task. In M. Mager, A. Ebrahimi, A. Oncevay, E. Rice, S. Rijhwani, A. Palmer, and K. Kann, editors, *Proceedings of the Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP)*, pages 156–162, 2023. doi: 10.18653/v1/2023.americasnlp-1.16.

C. Arnett and B. K. Bergen. Why do language models perform worse for morphologically complex languages? *arXiv preprint arXiv:2411.14198 [cs.CL]*, 2024. URL `http://arxiv.org/abs/2411.14198`.

E. Asgari, Y. E. Kheir, and M. A. S. Javaheri. MorphBPE: A morpho-aware tokenizer bridging linguistic complexity for efficient LLM training across morphologies. *arXiv preprint arXiv:2502.00894 [cs.CL]*, 2025. URL `http://arxiv.org/abs/2502.00894`.

S. Higashiyama and M. Utiyama. Comprehensive evaluation on lexical normalization: Boundary-aware approaches for unsegmented languages. *arXiv preprint arXiv:2505.22273 [cs.CL]*, 2025. URL `http://arxiv.org/abs/2505.22273`.

J. F. Hu. Tokenization strategies for low-resource agglutinative languages in Word2Vec: Case study on Turkish and Finnish. In *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pages 1–6, 2025. doi: 10.1109/acdsa65407.2025.11165911.

Meta AI. facebook/nllb-200-distilled-600M. Hugging Face model card, 2022. URL `https://huggingface.co/facebook/nllb-200-distilled-600M`.

NLLB Team, M. R. Costa-jussà, J. Cross, O. Çelebi, M. Elbayad, K. Heafield, K. Heffernan, E. Kalbassi, J. Lam, D. Licht, J. Maillard, A. Sun, S. Wang, G. Wenzek, A. Youngblood, B. Akula, L. Barrault, G. M. Gonzalez, P. Hansanti, and J. Hoffman. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672 [cs]*, 2022. URL `https://arxiv.org/abs/2207.04672`.

K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu. BLEU: A method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2002. doi: 10.3115/1073083.1073135.

M. Popović. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT)*, 2015. doi: 10.18653/v1/w15-3049.

M. Popović. chrF++: Words helping character n-grams. In *Proceedings of the Second Conference on Machine Translation (WMT)*, Copenhagen, Denmark, 2017. URL `https://aclanthology.org/W17-4770/`.

M. Post. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation (WMT)*, 2018. doi: 10.48550/arXiv.1804.08771.

A. Rios. *Applying Finite-State Techniques to a Native American Language: Quechua*. Licentiate thesis, University of Zurich, 2010. URL `https://www.cl.uzh.ch/dam/jcr:00000000-6a77-a254-0000-000009614c8d/rios_liz_applying_finite_state.pdf`.

R. Sennrich, B. Haddow, and A. Birch. Improving neural machine translation models with monolingual data. *arXiv preprint arXiv:1511.06709 [cs.CL]*, 2016. URL `http://arxiv.org/abs/1511.06709`.

SomosNLP Hackathon 2022. spanish-to-quechua. Hugging Face Datasets, 2022. URL `https://huggingface.co/datasets/somosnlp-hackathon-2022/spanish-to-quechua`.

I. Vieira, W. Allred, S. Lankford, S. Castilho, and A. Way. How much data is enough data? fine-tuning large language models for in-house translation: Performance evaluation across multiple dataset sizes. *arXiv preprint arXiv:2409.03454 [cs.CL]*, 2024. URL `http://arxiv.org/abs/2409.03454`.